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(54) ROTARY AND/OR SWIVEL UNIT

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(57) **ABSTRACT**

A rotary and/or swivel unit for attachment to a construction machine contains a connection part and a support arranged movable on the connection part for a tool or a quick changer for coupling a tool or attachment. The support is arranged on the connection part so as to be rotatable by a motor around an axis of rotation by means of a rotary drive and/or tiltable around a pivot axis by means of a pivot drive. The rotary and/or swivel unit has a central lubricant supply with an automatically actuated lubricant pump provided with a lubricant reservoir for delivering lubricant to different lubricating points.

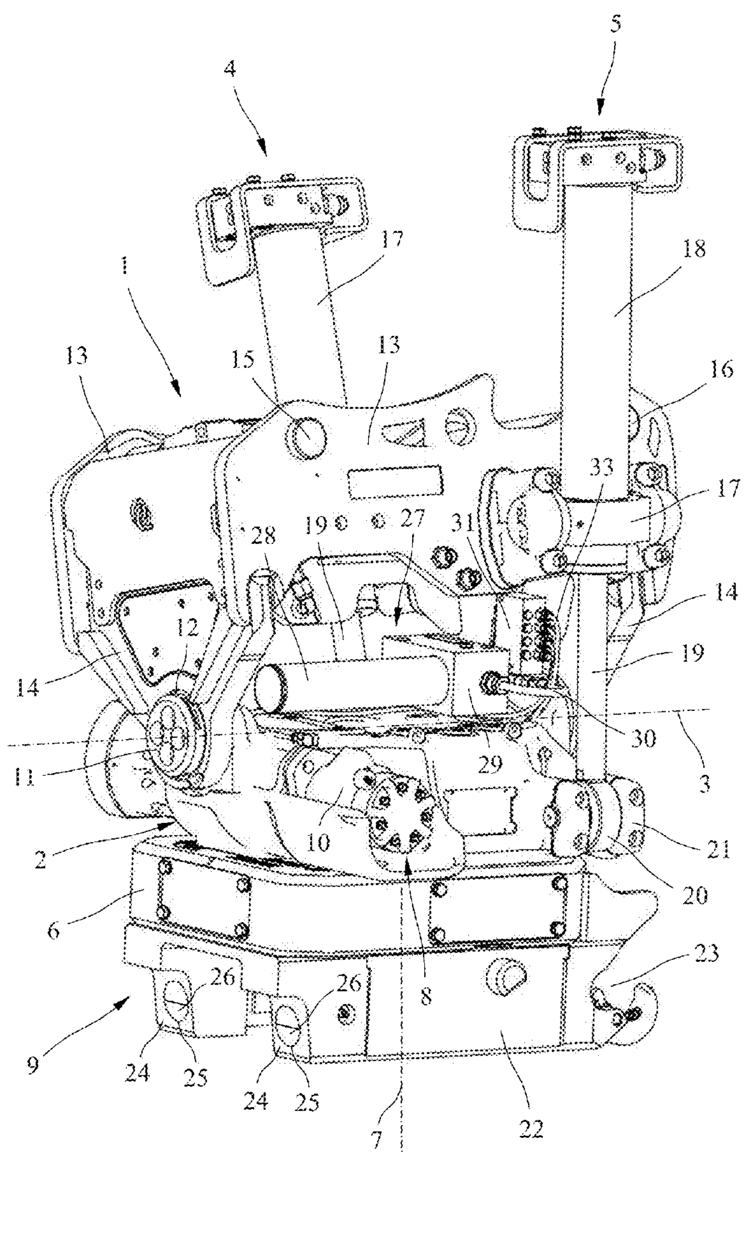

Fig. 1

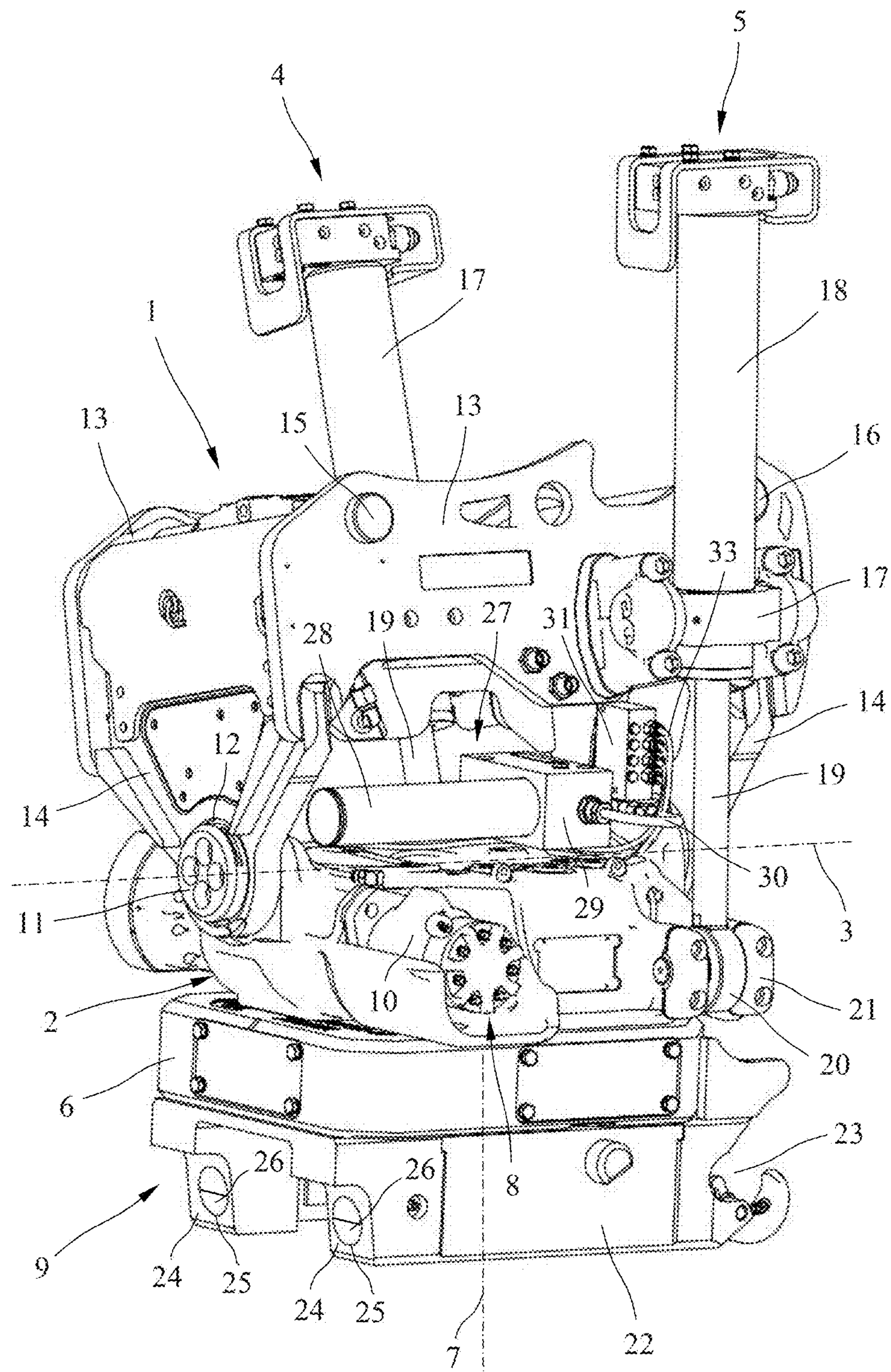
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Fig. 2

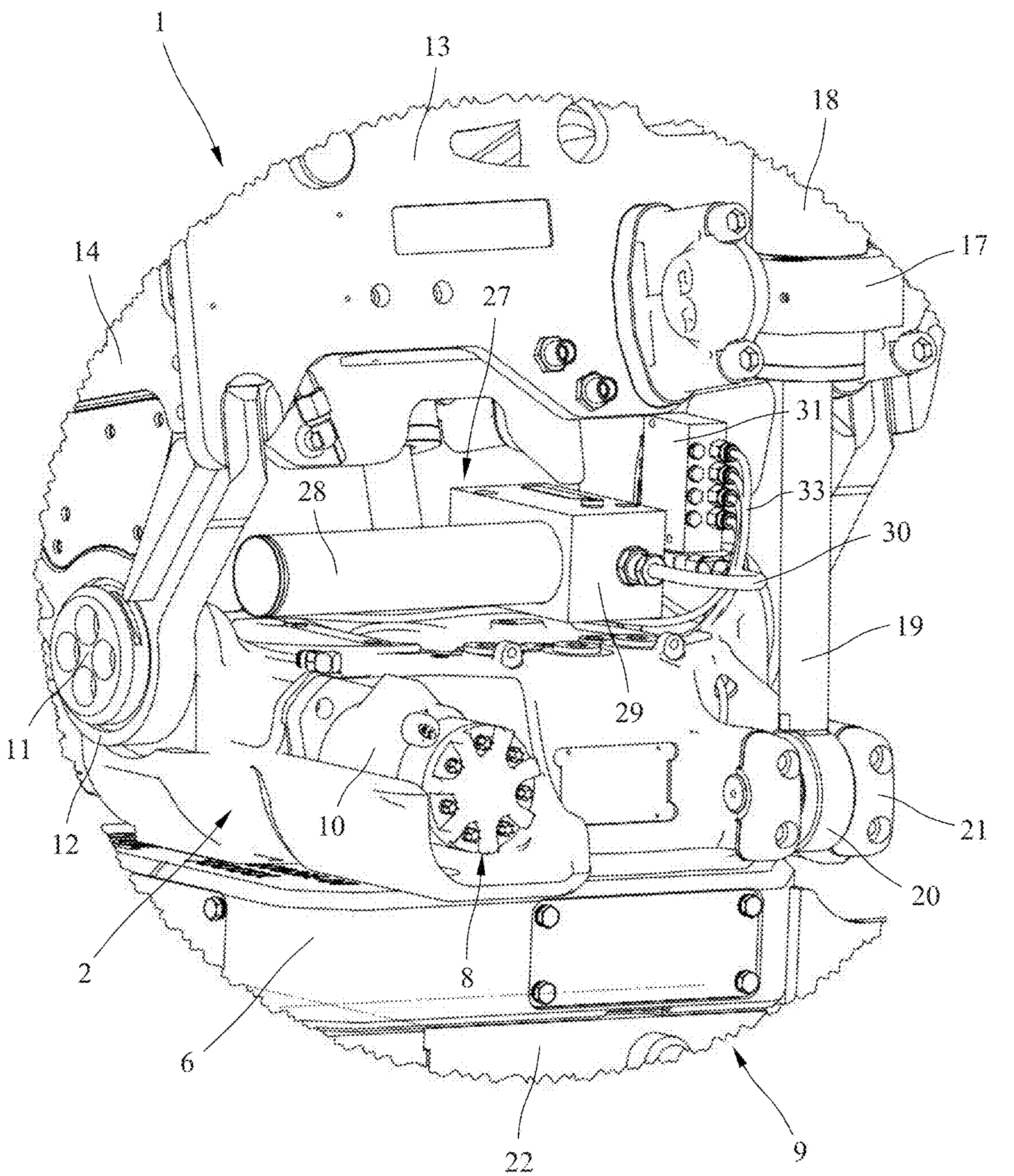
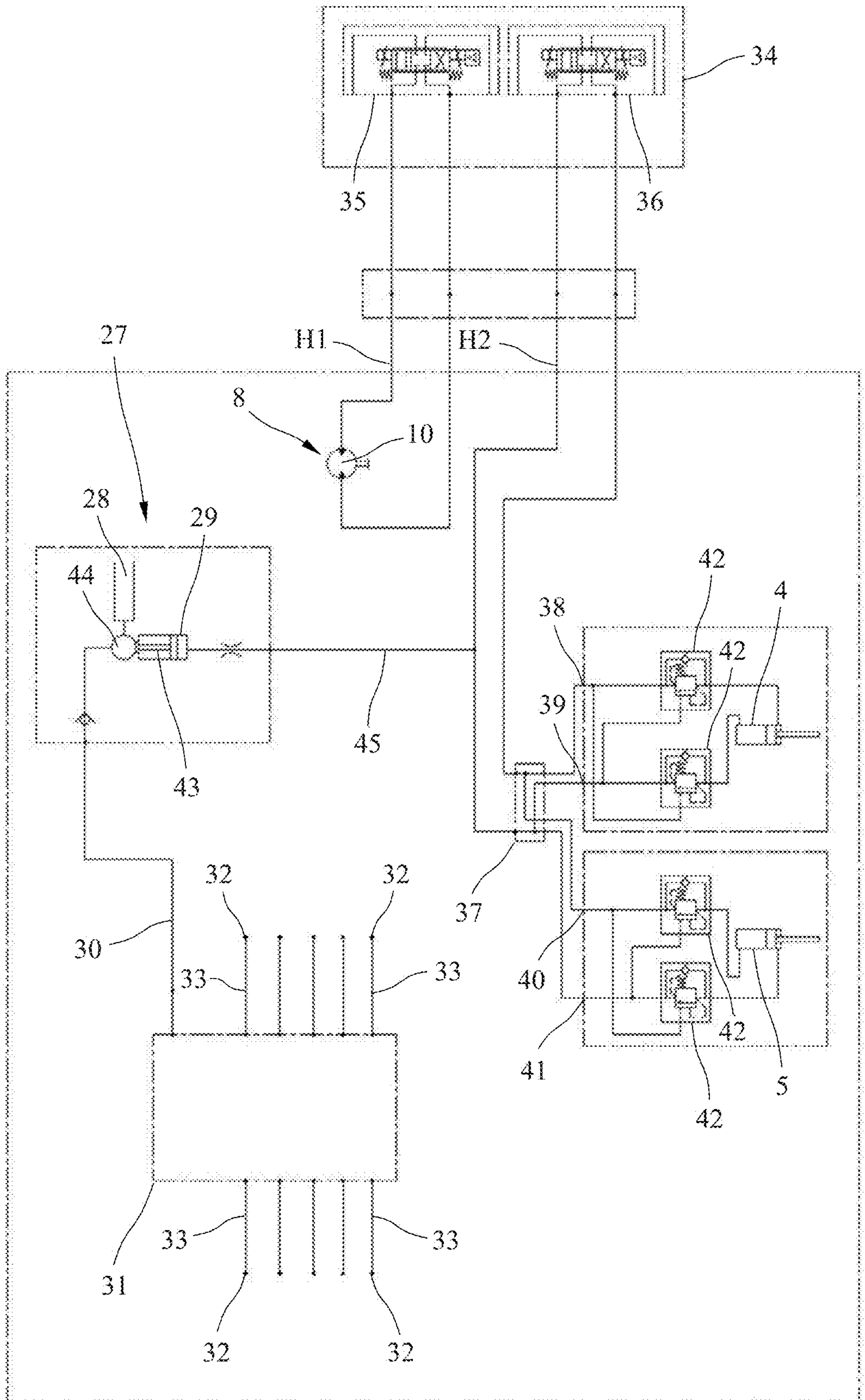


Fig. 3



ROTARY AND/OR SWIVEL UNIT

FIELD OF THE INVENTION

[0001] The invention relates to a rotary and/or swivel unit for attachment to a construction machine.

BACKGROUND OF THE INVENTION

[0002] Such rotary and/or swivel units are used on excavators or other comparable construction machines between an attachment of the construction machine and the tool as a rotary and/or swivel unit to expand the range of movement. Such rotary and swivel units ordinarily contain a connection part mountable on the construction machine and a support arranged movable on the connection part for coupling a tool or attachment in which the support is arranged on the connection part so as to be rotatable by a motor around an axis of rotation by means of a rotary drive and/or tiltable by means of a pivot drive around a pivot axis orthogonal to the axis of rotation. A combined rotary and/or swivel unit also referred to as a tiltrotator is also known from DE 10 2020 127 313 B3. In this case, the support of a quick changer is arranged rotatable around an axis of rotation by means of a hydraulic rotary drive and pivotable around a pivot axis orthogonal to the axis of rotation by means of a hydraulic pivot drive on a connection part mountable on an excavator. Such a tiltrotator enables the attachments connected to the quick changer, for example, tilting buckets, grapples, shears, compactors, magnets, hydraulic hammers or the like, to be rotated not only around a pivot axis arranged transversely the longitudinal axis of an excavator boom, but also around an axis of rotation orthogonal to this pivot axis.

[0003] Rotary and/or swivel units of the aforementioned type ordinarily have a number of different lubricating points, which must be lubricated regularly to ensure proper operation and to avoid wear. However, regular lubrication during daily operation is often neglected or forgotten, which can lead to undesired wear and the related damage and disruptions in operation.

SUMMARY OF THE INVENTION

[0004] One aspect of the invention relates to a rotary and/or swivel unit of the aforementioned type, which ensures regular lubrication of the different lubricating points even without a separate intervention.

[0005] Useful embodiments and advantageous modifications of the invention are disclosed herein.

[0006] The rotary and/or swivel unit according to the invention contains a connection part mountable on a construction machine and a support arranged movable on the connection part for connecting a tool or attachment, in which case the support is arranged on the connection part so as to be rotatable by a motor around an axis of rotation by means of a rotary drive and/or tiltable around a pivot axis by means of a pivot drive. The rotary and/or swivel unit also contains a central lubricant supply with an automatically actuated lubricant pump provided with a lubricant reservoir. The automatically actuated lubricant pump arranged on the rotary and/or swivel unit enables the different lubricating points of the rotary and/or swivel unit to be supplied with a lubricant provided in a lubricant reservoir even without a lubricant connection to a central lubricating system on the construction machine and without any separate intervention by the operator. the lubricant pump can be activated automatically, for example, when the rotary and/or swivel drive is actuated or when the construction machine is started. This means that the lubrication of the different lubricating points, which is essential for preventing wear, cannot be overlooked. Owing to the fact that the lubricant supply to the rotary and/or swivel unit is independent of the central lubricating system of the construction machine and integrated into the rotary and/or swivel unit, no lubricant lines and specially designed couplings are necessary between the rotary and/or swivel unit and a central lubricating system provided on the construction machine.

[0007] In a particularly useful variant, a housing part is arranged on a connection part, which is mounted so as to be pivotable around a pivot axis and tiltable by means of a pivot drive in which the support is rotatably mounted around an axis of rotation orthogonal to the pivot axis, and which can be rotated by a motor by means of the rotary drive. Such a rotary and swivel unit, referred to as a tiltrotator, enables the attachments connected to the rotary and swivel unit, for example, tilting buckets, grapples, shears, compactors, magnets, hydraulic hammers or the like, to be rotated not only around the pivot axis arranged transversely to the longitudinal axis of the excavator boom, but also around an axis of rotation orthogonal to this pivot axis.

[0008] In a particularly useful variant, the lubricant pump is connected via a supply line to a distributor to distribute the lubricant delivered by the lubricant pump to the different lubricating points. This enables the lubricant to reach the corresponding lubricating points in a predetermined manner. The amount of lubricant that the individual lubricating points require can be stipulated via the distributor unit. Several lubricant supply lines leading to the different lubricating points are connected to the distributor unit.

[0009] A lubricant amount regulation device for adjusting the amount of lubricant delivered by the lubricant pump when same is activated is expediently arranged on the lubricant pump. This makes it possible to adjust the amount of lubricant delivered by the lubricant pump from the lubricant reservoir when the pump is activated.

[0010] The lubricant pump can be arranged on or adjacent to the housing part of the rotary and/or swivel unit tiltable around the pivot axis. The distributor unit is also preferably arranged on or adjacent to the housing part tiltable around the pivot axis.

[0011] In a particularly advantageous variant, the lubricant pump can be designed as a hydraulically operated lubricant pump with a hydraulically operated pump piston. However, the lubricant pump can also be electrically operated and activated, for example, by an electric control when the construction machine is started. Regular lubrication of the rotary and/or swivel unit can also thereby be ensured.

[0012] The lubricant pump can expediently be activated by controlling the pivot drive or by controlling the rotary drive. In a hydraulic rotary and pivot drive and a hydraulically operating lubricant pump, the lubricant pump can be connected via a bypass line to a hydraulic circuit H**2** for controlling the pivot drive or a hydraulic circuit H**1** for controlling the rotary drive.

[0013] The rotary drive can include a hydraulic motor that can be controlled by the first hydraulic circuit H**1** and the pivot drive can be formed by two double-acting actuating cylinders movable in opposite directions and controllable by the second hydraulic circuit H**2**.

[0014] A quick changer designed for connecting a tool can be arranged on the support. However, a grapple or other tool can also be directly arranged on the support. The support can also be designed as the housing of a quick changer.

[0015] The connection part can contain two spaced-apart pin-like coupling elements for coupling the connection part to a quick changer. The connection part and thus the rotary and/or swivel unit can therefore be connected via a quick changer to an excavator boom and used in a sandwich assembly with an additional lower quick changer for automatically changing attachments. However, the connection part can also be permanently attached to an excavator arm.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Further details and advantages of the invention will become apparent from the following description of preferred embodiments with reference to the drawing. In the drawing:

[0017] FIG. **1** shows a rotary and swivel unit with a quick changer and an automatic lubricant supply in a first perspective view;

[0018] FIG. **2** shows an enlarged detailed view of the rotary and swivel unit of FIG. **1**; and

[0019] FIG. **3** shows a hydraulic circuit arrangement for controlling the automatic lubricant supply of the rotary and swivel unit shown in FIGS. **1** and **2**.

DETAILED DESCRIPTION

[0020] The rotary and swivel unit shown in a perspective view in FIG. **1** and also referred to as a tiltrotator contains a connection part **1** mountable either directly or via a hydraulically operated quick changer on an excavator boom or another attachment of a construction machine, on which a housing part **2** is mounted to pivot around the pivot axis **3** and arranged tiltable around the pivot axis **3** by means of a pivot drive which, in this case, is formed by two hydraulic actuating cylinders **4** and **5**. In the housing part **2** tiltable relative to connection part **1**, a support **6** is rotatably mounted around an axis of rotation **7** orthogonal to the pivot axis **3** and can be rotated 360° by means of a rotary drive **8**. In the embodiment shown, a quick changer **9** designed for connecting a tool is arranged on the support **6**. However, a grapple or other tool can be directly arranged on the support **6**. In addition, the support **6** can also be designed as the housing of a quick changer.

[0021] The rotary drive **8**, which is designed here as a hydraulic drive, contains a hydraulic motor **10** by means of which the support **6** can be rotated by a motor around the axis of rotation **7**. by 360° relative to the housing part **2**, via a gear mechanism with a worm gear arranged inside the housing part **2** and a worm drive rotatable by the hydraulic motor **10**.

[0022] The housing part **2** is arranged on connection part **1** via laterally protruding journals **11** inside two spaced-apart bearing lugs **12** so as to be pivotable around the pivot axis **3**, which is orthogonal to axis of rotation **7**, and can be pivoted by approximately +/−45° around the pivot axis **3**, relative to the connection part **1**, via the hydraulic pivot drive formed by the two actuating cylinders **4** and **5**. The attachments connected to the quick changer **9** can not only be rotated around the axis of rotation **7** by the tiltrotator which is also referred to as a rotary-pivot drive, but also tilted relative to connection part **1** around the pivot axis **3** orthogonal to axis of rotation **7**, thereby expanding the range of movement increasing the area of application.

[0023] In the embodiment shown, the connection part **1** has two parallel side walls **13** as well as front and rear crosspieces **14** with the bearing lugs **12** arranged thereon. The housing part **2** is mounted to pivot around the pivot axis **3** via the two journals **11** inside the bearing lugs **12** of the front and rear crosspieces **14**. Two parallel and spaced-apart pin-like coupling elements **15** and **16** for connecting to a hydraulically operated quick changer (not shown here) arranged on the construction machine are arranged in the two side walls **13** of the connection part **1** shown.

[0024] An embodiment of a hydraulically operable quick changer arranged on an excavator arm or other construction machine for automatically connecting the tiltrotator is disclosed in DE 10 2018 128 479 A1. This document is explicitly referred to with regard to the known design and method of operation of such a hydraulically operable quick changer. However, the connection part **1** could also be directly mounted on a boom and a coupling of an excavator without the intermediate connection of a hydraulically operated quick changer.

[0025] In the embodiment shown, the pivot drive is formed by the double-acting actuating cylinders **4** and **5**, each having a cylinder housing **18** fastened via a mount **17** to the corresponding side wall **13** of the connection part **1** and a piston rod **19** slidably arranged and hydraulically movable in the cylinder housing **18**. The free ends of the piston rods **19** are connected to the drive housing **9** via a joint eye **20** and a corresponding mount **21**. By retracting and extending the two piston rods **19** of the actuating cylinders **4** and **5**, the housing part **2** with the support **6** rotatably mounted therein can thus be tilted relative to the connection part **1**. The pivot drive, however, can also be formed by a hydraulic motor or another appropriate hydraulic drive.

[0026] The quick changer **9** designed for coupling an attachment or tool contains a housing **22** designed as a welded structure or as a cast part, which has first receptacles **23** open to one side for accommodating and holding a first pin-like coupling element on the tool side on one side, as well as second receptacles **24** open toward the bottom for accommodating and holding a second pin-like coupling element on the tool side on the other side. The first receptacles **23** open to one side are designed as claw- or fork-like. The second receptacles **24** to the other side toward the bottom have a curved lower support surface for supporting a pin-like coupling element. A manually or hydraulically operated locking device with two locking pins **26** movable inside holes **25**, between an extended locking position and a retracted unlocking or change position, is provided on the second receptacles **24**. The locking pins **26** can be moved manually via a manually pivotable lever or hydraulically between the retracted unlocking or change position and the extended locking position. Various tools or attachments can be connected to the quick changer **9**, for example, tilting buckets, grapples, shears, compactors, magnets, hydraulic hammers or the like.

[0027] As can be seen in particular from the enlarged detailed view of FIG. **2**, the rotary and swivel drive contains a central lubricant supply **27** with an automatically operable lubricant pump **29** provided with a lubricant reservoir **28** and a distributor unit **31** connected to the lubricant pump **29** via a supply line **30** for distributing the lubricant to different lubricating points **32**. The lubricant reservoir **28**, which is designed as a container for accommodating a sufficient amount of grease or a corresponding lubricant, is arranged on the lubricant pump **29** and can be mounted together with it on the housing part **2**. A lubricating grease can be used as the lubricant, which is supplied to the lubricant reservoir **28**. Several lubricant supply lines **33** leading to the individual lubricating points **32** are connected to the distributor device **31** for distributing the lubricant to the different lubricating points **32**. The amount of lubricant required by the individual lubricating points **32** can be stipulated via the distributor unit **31**.

[0028] A circuit diagram of a hydraulic circuit arrangement for controlling the rotary drive **3** formed by the hydraulic motor **10**, the pivot drive formed by the two actuating cylinders **4** and **5** and the lubricant pump **29** of the central lubricant supply **27** of the rotary and swivel drive is shown in FIG. **3**. The circuit arrangement includes a control device **34** arranged, for example, on an excavator, which includes a first control valve **35** designed as a 4/3-way valve for controlling the first hydraulic circuit H**1** used to control the hydraulic motor **10** of the rotary drive **3**, and a second control valve **36** designed as a 4/3-way valve for controlling a second hydraulic circuit H**2** used to control cylinders **4** and **5** of the pivot drive. The two actuating cylinders **4** and **5** of the pivot drive are controlled by the second hydraulic circuit H**2** via branch **37** and four connections **38** to **41** so that the actuating cylinders **4** and **5** can be extended or retracted in opposite directions and the support **6** thus tilted around the pivot axis **3**. A retarding brake check valve **42** is arranged upstream of the connections for retracting and extending the two actuating cylinders **4** and **5**.

[0029] In the embodiment shown, the lubricant pump **29** is designed as a hydraulically operated lubricant pump with a hydraulically operated pump piston **43** and a lubricant amount regulation device **44** for adjusting the amount of lubricant delivered. The lubricant pump **29** is controlled by the hydraulic circuit H**2** via a bypass line **45** branching off from the hydraulic circuit H**2**. This ensures that when the two actuating cylinders **4** and **5** for operating the pivot drive are activated, the lubricant pump **29** is also simultaneously activated to supply the lubricating points **32** with lubricant. The bypass line **45**, however, can also branch off from the first hydraulic circuit H**1** so that when the hydraulic motor **10** of the rotary drive **8** is activated, the lubricant pump **29** is activated. Regular lubrication can also thereby be ensured.

[0030] The invention is not limited to the rotary and swivel unit described in detail above. A central and automatically actuated lubrication device can also be used for pure rotary or swivel units.

LIST OF REFERENCE NUMBERS

[0031] **1** Connection part
[0032] **2** Housing part
[0033] **3** Pivot axis
[0034] **4** First actuating cylinder
[0035] **5** Second actuating cylinder
[0036] **6** Support
[0037] **7** Pivot axis
[0038] **8** Rotary drive
[0039] **9** Quick changer
[0040] **10** Hydraulic motor
[0041] **11** Journal
[0042] **12** Bearing lug
[0043] **13** Side wall
[0044] **14** Crosspiece
[0045] **15** First coupling element
[0046] **16** Second coupling element
[0047] **17** Mount
[0048] **18** Cylinder housing
[0049] **19** Piston rod
[0050] **20** Joint eye
[0051] **21** Mount
[0052] **22** Housing
[0053] **23** First receptacle
[0054] **24** Second receptacle
[0055] **25** Hole
[0056] **26** Locking pin
[0057] **27** Central lubricant supply
[0058] **28** Lubricant reservoir
[0059] **29** Lubricant pump
[0060] **30** Supply line
[0061] **31** Distributor unit
[0062] **32** Lubricating point
[0063] **33** Lubricant supply line
[0064] **34** Control device
[0065] **35** First control valve
[0066] **36** Second control valve
[0067] **37** Branch
[0068] **38** First connection
[0069] **39** Second connection
[0070] **40** Third connection
[0071] **41** Fourth connection
[0072] **42** Retarding brake check valve
[0073] **43** Pump piston
[0074] **44** Lubricant amount regulation device
[0075] **45** Bypass line

1. A rotary and/or swivel unit for attachment to a construction machine, which contains a connection part and a support arranged movable on the connection part for a tool or a quick changer for coupling a tool or attachment, in which the support is arranged on the connection part so as to be rotatable by a motor around an axis of rotation by means of a rotary drive and/or tiltable around a pivot axis by means of a pivot drive, wherein a central lubricant supply with an automatically actuated lubricant pump is provided with a lubricant reservoir for delivering a lubricant to different lubricating points of the rotary and/or swivel unit.

2. The rotary and/or swivel unit according to claim **1**, wherein a housing part is arranged on the connection part, which is mounted so as to be pivotable around the pivot axis and tiltable by the pivot drive in which the support is rotatably mounted around an axis of rotation orthogonal to the pivot axis, and which can be rotated by a motor by means of the rotary drive.

3. The rotary and/or swivel unit according to claim **1**, wherein the lubricant pump is connected via a supply line to a distributor unit to distribute the lubricant delivered by the lubricant pump to the different lubricating points.

4. The rotary and/or swivel unit according to claim **3**, wherein the lubricant supply lines leading to the different lubricating points are connected to the distributor unit.

5. The rotary and/or swivel unit according to claim **1**, wherein a lubricant amount regulation device is arranged on the lubricant pump for adjusting the amount of lubricant delivered by the lubricant pump when the lubricant pump is activated.

6. The rotary and/or swivel unit according to claim **2**, wherein the lubricant pump is arranged on the housing part.

7. The rotary and/or swivel unit according to claim **1**, wherein the lubricant pump is designed as a hydraulically operated lubricant pump with a hydraulically operated pump piston.

8. The rotary and/or swivel unit according to claim **1**, wherein the lubricant pump is activated by controlling the pivot drive or by controlling the rotary drive.

9. The rotary and/or swivel unit according to claim **8**, wherein the lubricant pump is connected via a bypass line to a pivot drive hydraulic circuit for controlling the pivot drive, a rotary hydraulic circuit for controlling the rotary drive or another hydraulic circuit.

10. The rotary and/or swivel unit according to claim **9**, wherein the rotary drive includes a hydraulic motor that can be controlled by the rotary drive hydraulic circuit.

11. The rotary and/or swivel unit according to claim **9**, wherein the pivot drive is formed by two double-acting actuating cylinders movable in opposite directions and controllable by the pivot drive hydraulic circuit.

12. The rotary and/or swivel unit according to claim **1**, wherein a quick changer designed for connecting a tool or attachment is arranged on the support.

13. The rotary and/or swivel unit according to claim **1**, wherein the connection part contains two spaced-apart pin-like coupling elements for coupling the connection part to a quick changer.

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